COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

B.Tech Programme in ELECTRONICS ENGINEERING (A batch)

First Internal Examination, April 2024

Second semester (2023 scheme)

23-200-0201 *LINEAR ALGEBRA & TRANSFORM TECHNIQUES*

Time : 2 hours Max.Marks : 30

Course Outcome (CO)

On completion of this course the student will be able to

CO1. Solve linear system of equations and to determine Eigen values and vectors of a matrix

CO2.Exemplify the concept of vector space and sub space.

CO3.Determine Fourier series expansion of functions and transform.

CO4. Solve linear differential equation and integral equation using Laplace transform.

PART A

(Answer all questions) Max.marks

1. Find the rank of the matrix $\begin{bmatrix} 1 & 3 & 5 \\ 2 & -1 & 4 \\ -2 & 8 & 2 \end{bmatrix}$ 2 [co1] [po1] L2

2. Find the eigen values of $\begin{bmatrix} 6 & 3 \\ 6 & -1 \end{bmatrix}$ 2 [co1] [po1] L1

3. Apply Cayley-Hamilton theorem to find A^{-1} if

   A = $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ 2 [co1] [po1] L3

4. Prove that the eigen values of a triangular matrix are just the diagonal elements of the matrix. 2 [co1] [po1] L2

5. For what values of λ and μ, the system of equations

   $x+2y+3z=4$, $x+3y+4z=5$, $x+3y+\lambda z=\mu$

   have no solution. 2 [co1] [po1] L1

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